

Electrical Testing & Commissioning Checklist

A field-ready checklist for pre-commissioning, energisation and handover of LV / MV / HV industrial electrical systems. Use this as the working spine for your T&C cycle — every item produces a documented test record.

Version 1.0 · Aligned with IS/IEC/CEA practices

1. Pre-Commissioning — Documentation Pack

- Approved Single Line Diagram (SLD) and protection coordination study on site
- Cable schedule, routing drawing and termination details verified
- Equipment GA, foundation and earthing layout drawings released for construction
- OEM type test and routine test reports collected
- Relay setting calculation sheet approved by consultant / utility
- Method of Statement and HIRA prepared and signed off

2. Visual & Mechanical Inspection

- Transformer / switchgear installed plumb, level and torqued to OEM values
- Busbar joints torque-checked; tightness markings applied
- Cable glanding, lugs, ferrules and phase identification verified
- Earthing continuity to body, neutral, LA and grid confirmed
- Anti-condensation heaters, space heaters and thermostats operational
- Nameplate data matches design documents and SLD

3. Electrical Tests — LV / MV / HV

- IR test on cables, motors and transformers — phase-phase / phase-earth
- Polarisation Index (PI) and DAR for rotating machines
- HV withstand test (AC up to 70 kV / DC up to 80 kV) per IS/IEC schedule
- Transformer ratio (TTR), magnetic balance, vector group and winding resistance
- Tan-delta and capacitance on bushings, CTs, PTs and cables
- CT/PT polarity, ratio and burden; secondary injection on metering circuits

 Earth electrode resistance and earth grid impedance measurement

4. Protection & Control Verification

- Numerical relay configuration uploaded and locked against approved settings
- Primary and secondary injection of all protection elements (50, 51, 87, 64, 27, 59)
- Trip / close circuits proven end-to-end including DC supply healthiness
- Interlock matrix tested — mechanical, electrical and SCADA permissives
- Annunciation, event logging and SNTP time-sync verified
- Backup protection coordination time-graded with upstream / downstream

5. Auxiliaries & Control Power

- Battery bank capacity discharge test; float / boost charger function check
- UPS / inverter changeover and battery autonomy verified
- AC and DC distribution boards tested for selectivity and earth fault
- SCADA / RTU communication, IEC 61850 GOOSE and DNP3 mapping confirmed

6. Energisation Readiness

- Statutory approval from Electrical Inspectorate / CEIG obtained
- Permit-to-work, LOTO, PPE and emergency response plan in place
- Pre-charge inspection by client, EPC and VECTREV engineer signed
- Phase sequence and synchronisation on incomers and tie-breakers
- First charge no-load observation for minimum 1 hour documented
- Load build-up plan and shutdown criteria agreed with operations

7. Handover Documentation

- Test reports compiled, indexed and signed by all parties
- As-built drawings updated and issued in editable + PDF formats
- Spare parts list, special tools and OEM warranty certificates handed over
- Operator training conducted; attendance and competency records filed
- Punch-list closed-out with photographic evidence

Need a VECTREV engineer to run this on your site?

Call **+91 88796 08428** or email **contact@vectrev.in**. An engineer — not a salesperson — responds within 24 hours.